

EDUCATION

University of New Mexico, Albuquerque, NM

Fall 2026

Bachelor/Masters of Science: Computer Science

Relevant Coursework: Data Structures and Algorithms I & II, Machine Learning, Computer Operating Systems, Computer Architecture, Computer Networks, Data Organization, Design of Large Programs, Mobile Computing, Parallel Processing

TECHNICAL SKILLS & CERTIFICATIONS

SKILLS: Java, Python, C, React Native, JavaScript, Haskell, Delphi, SQL, Scikit-Learn, Algorithm Analysis, Support Vector Machines, Linear/Nonlinear Regression, Classification, Photogrammetry, Computer Vision, Xcode, Swift, Docker

CERTIFICATIONS: [Python for Data Science, AI & Development](#), IN PROGRESS: : UNM Lobo Venture Capitalist Cup

HIGHLIGHTED PROJECTS AND WORK EXPERIENCE

Square Scan – Personal Project

May 2025 - Present

Mobile Photogrammetry & Measurement System

- Built a React Native + iOS image capture app to measure residential exterior wall planes using photogrammetry and computer vision for use by contractors to estimate square footage and quote prices.
- Reconstructed a scaled 3D point cloud from a given building image set through AWS to run GPU accelerated jobs via EC2/Batch and Redis for fast queuing. Implemented a RANSAC algorithm to detect wall planes of the point cloud and a YOLOv8-Segmentation model to detect and subtract windows/doors to compute final per-plane square footage.
- Presented application and achieved first place for Judge's Choice in the University of New Mexico Pitch Competition.

2025 Lobo Hackathon - Winner, University of New Mexico

Apr 2025

- Co-built Lobo Library, a prototyped mobile app unifying the cumbersome library services (scheduling, checkouts, real-time availability) into a streamlined, student-friendly interface using React Native.
- Liaised with library and campus IT systems stakeholders to gather information and develop features that are in demand by student users and in compliance with current technical protocols.
- Delivered a final pitch to a panel of industry judges, selected as one of 9 winning teams for execution and potential.

Intro to Machine Learning, University of New Mexico

Feb 2025

Scalability of Support Vector Machines

- Implemented a custom *LinearSVC* class in Python to train a linear Support Vector Classifier with soft-margin optimization using hinge loss and L2 regularization.
- Designed a function to generate linearly separable datasets based on a random separation hyperplane.
- Benchmarked model performance on self-created datasets; analyzed trade-offs between scikits-learn's LinearSVC.

Technolution Software, Bad Salzufen, Germany

Summer 2024

PLC Programmer

- Created editors for the parameters of machinery software that execute precision cuts in industrial production.
- Connected editors by executing SQL queries to company relational databases.
- Contributed to Delphi code production by creating various database table editors used in requested client's software.

Intermediate Programming, UNM

Jul 2023

Concurrent Processing Over a Network

- Developed a Java client-server system utilizing concurrent processing to perform matrix addition
- Established TCP sockets and data streams enabling client-server communication
- Designed server to split the matrix addition across four threads, enabling faster computation

LEADERSHIP & PROFESSIONAL DEVELOPMENT EXPERIENCE

Thrive Scholars - Scholar

Apr 2020-Present

- Selected as one of 240 applicants for a 6-year college access, college success, and professional development program for high-achieving, underrepresented first-generation talent.
- Participated in a four-year comprehensive Career Development Program that includes professional Career Coaching, case studies, and projects to hone analytical, quantitative, and communication skills.

University of New Mexico - IT Network Student Technician

Aug 2024-Present

- Resolve tickets regarding static IP allocation, port activation, and network config across campus infrastructure.